How accurately can task performance be predicted in real time using machine learning models trained on multimodal physiological signals collected during virtual reality pilot training tasks?

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Project Definition

Training professionals in high-stake environments requires the acquisition of complex psychomotor and cognitive skills and attaining expert-level competency proves challenging due to strict safety concerns and constrained training resources (Nassar et al., 2021). In the aviation sector, the application of machine learning in pilot training remains largely underexplored (Caballero et al., 2023). Machine learning techniques leveraging physiological signals from trainee pilots incorporate more readily into legacy training equipment (Caballero et al., 2023); moreover, they enable precise error measurement, replacing labor-intensive evaluation processes typically performed by human instructors and allowing those instructors to allocate more time and attention to individualized trainee development (Barry et al, 2025). This project seeks to investigate real-time task performance prediction by utilizing machine learning and multimodal physiological data, with the findings potentially supporting the further development of adaptive and scalable training systems.

Motivation

Flight simulators are becoming increasingly important for training and research due to growing concerns about flight safety, the shortage of pilots, and the rapid advancements in the aviation industry (Wang, 2024). Virtual reality has advanced traditional pilot training by integrating VR headsets into simulators and enhancing visual realism through high-fidelity 3D flight scenarios. This integration provides trainees with an immersive and practical experience within authentic virtual cockpits, thereby improving the realism of training, cost-effectiveness, operational safety, and overall training efficiency (Wang, 2024). However, existing simulator-based training depends extensively on human instructors and evaluators for performance monitoring and adaptive difficulty adjustment, thereby constraining training output amid shortage of instructors (Caballero et al., 2023). Adaptive automated training systems that continuously monitor performance and dynamically adjust task difficulty may therefore alleviate the negative effects of limited instructor availability on training duration and the operational demands placed on working pilots (Barry et al., 2025).

Electroencephalograms (EEG) and other psychophysiological data have been employed to track pilots' performance in real time and to forecast fatigue (Alreshidi et al., 2024; Dideriksen & Williams, 2019; Wilson et al., 2019). Many of these existing studies have investigated pilot performance either in simulator cockpits with 2D displays or in real-flight scenarios. However, performance monitoring and modeling within immersive VR training environments remain comparatively underexplored. Some studies have demonstrated the performance of machine learning models using single-modal signals to analyze pilot performance in VR-integrated simulators (Moore et al., 2023; Rajendran et al., 2021). Thus, further research is still needed to explore machine learning approaches for assessing training performance using multimodal physiological signals within integrated virtual reality environments, and how these models can be used to support real-time adaptive feedback.

Research Question

How accurately can task performance be predicted in real time using machine learning models trained on multimodal physiological signals collected during virtual reality pilot training tasks?

Background

Several studies have explored diverse machine learning models using different physiological signals to aid in developing an adaptive VR-integrated flight simulator for pilot training. A commonly used dataset in relevant research is the CogPilot dataset (Rao et al., 2022), which contains multimodal physiological, eye-tracking, and simulator data from 35 participants performing simulated landing tasks in a VR-integrated flight simulator. Caballero et al. (2023) utilized this dataset for a flight-difficulty classification task. They trained a machine learning pipeline on multimodal physiological signals from 21 participants. An AdaBoost-based model, using features derived through functional Principal Component Analysis and feature selection, achieved approximately 87% accuracy and an Area Under the Curve (AUC) of 0.88 (Caballero et al., 2023). Using the same dataset, Moore et al. (2023) focused on eye-tracking and electrodermal activity signals for the same classification task, combining the Minimally Random Convolutional Kernel Transform with a deep neural network. Their model achieved an AUC of 0.91 while reducing the original time-series data by approximately 99.7%. Barry et al. (2025) also utilized this dataset to develop a machine learning pipeline predicting participant landing error from multimodal physiological and aircraft data. They first transformed high-frequency time-series data into tabular features using summary statistics and functional principal component analysis, then applied BorutaSHAP for feature selection and trained several tree-based models within their multimodal functional learning framework for error regression. These studies provide a foundation for automated flight difficulty classification and task performance prediction to support intelligent instructors and evaluator systems. However, they primarily rely on post-hoc analysis rather than continuous, real-time prediction. Thus, further research is needed to examine machine learning approaches for online performance prediction.

Several other studies have also investigated the relationship between physiological signals and cognitive load during VR flight tasks. Koerner et al. (2025) investigated heart rate as an index of mental workload and performance using the CogPilot dataset (Rao et al., 2022). They found that mean heart rate increased with task difficulty, decreased across repeated runs of the same difficulty, and rose near touchdown compared to the flight start, while correlating positively with mental workload and negatively with task performance. These results suggest that heart rate can serve as a practical signal for real-time task difficulty adjustment to maintain trainees within an optimal workload band (Koerner et al., 2025). Another study by Van Weelden et al. (2024) compared EEG-based workload (beta-ratio power) and flight performance between 2D Desktop and 3D VR flight simulations using the PC-7 simulator. 52 participants (university students) with no prior experience performed two tasks (speed change, medium turn) in a within-subjects designed experiment (Van Weelden et al., 2024). The results revealed that VR flight simulations produced greater task engagement as evidenced by elevated EEG beta-ratios compared to Desktop simulations, but beta-ratios failed to predict flight performance in either condition (Van Weelden et al., 2024). Their findings showed that VR-integrated simulations improved training performance, though more research is needed to investigate other physiological signals as parameters for optimizing the training experience.

Dataset Description

The project will use the CogPilot dataset (Rao et al., 2022). The dataset contains synchronized physiological data, including electromyography (EMG), forearm acceleration (ACC), electrodermal activity (EDA), electrocardiography (ECG), respiration (RES), eye tracking (ETK), and photoplethysmography (PPG), with nominal sampling rates ranging from 128 to 1024 Hz. It also includes aircraft and pilot performance data collected from 35 participants performing virtual landing tasks using an HTC Vive Pro Eye headset integrated with X-Plane 11 flight simulation software. Noisy data will be removed, and physiological signals will be normalized prior to machine learning model training. The data are stored as CSV files per stream with a common timestamp and are publicly available via PhysioNet (Goldberger et al., 2000).

This dataset is well suited because it directly targets predicting flight performance and task difficulty from multimodal physiological responses. It provides well-documented signals with explicit difficulty manipulation and performance metrics and has already supported difficulty classification and error prediction.

Instrumentation and Material

This project does not involve new data collection or physical experimental apparatus. All analyses will be based on the publicly available dataset.

Statistics, Algorithms, and Software

The problem will be formulated as a supervised regression task predicting continuous flight performance measures from 5-second sliding windows of multimodal physiological time-series data. Statistical analysis and modeling will be conducted in Python using NumPy, Pandas, SciPy, scikit-learn, XGBoost, PyTorch, and sktime, with Matplotlib for visualization. Random Forest (baseline) and XGBoost will both use sliding-window transformations of physiological time-series data. These models were previously compared by Barry et al. (2025) for regression tasks on the CogPilot dataset, providing established benchmarks that this project extends to real-time continuous error prediction. A 1D Convolutional Neural Network (CNN) will serve as the third model, as recent studies have demonstrated promising results using CNNs for time-series forecasting tasks (Wibawa et al., 2022). Model performance will be evaluated using leave-one-subject-out cross-validation and reported as root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2).

Milestones and Plan

17 Feb – 1 Mar	Finalize research question, objectives, and proposal.
1 Mar– 14 Mar	Data cleaning, preprocessing.
15 Mar – 22 Mar	Implement and test feature preparation for selected modalities. create consolidated run-level feature matrix and targets.
23 Mar – 30 Mar	Train and evaluate Random Forest model with subject-wise cross-validation; establish baseline metrics.
31 Mar – 6 Apr	Develop and tune XGBoost regressors; compare performance and feature importance with baseline model
7 Apr – 14 Apr	Implement and tune a 1D CNN
15 Apr – 21 May	Synthesize results; refine planned analyses if needed; finalize and submit the thesis.

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